

Explainable-by-Design Models for Speech Emotion Analysis with HuBERT and Concept Bottlenecks

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Abstract

This project studies explainability for speech paralinguistic analysis using audio-only emotion recognition on MELD. A frozen HuBERT representation is first used as a black-box speech baseline. An explainable-by-design Concept Bottleneck Model (CBM) is then introduced: HuBERT embeddings are mapped to acoustic proxy concepts, and emotion classification is performed through these concepts rather than directly from the full embedding. Explanations are evaluated using concept-level SHAP scores, ablation-based faithfulness tests, and targeted concept interventions. The black-box HuBERT model achieves the strongest macro-F1, while the CBM loses predictive performance but provides inspectable intermediate concepts. The results show that concept bottlenecks can make speech emotion predictions more transparent, but also reveal the difficulty of grounding emotion explanations in automatically extracted acoustic descriptors.

Keywords: explainable AI, speech emotion recognition, HuBERT, Concept Bottleneck Model, MELD, SHAP

Repository: <https://github.com/parsa-tavakoli/explainable-speech-emotion-cbm/tree/main> (full results are available in the GitHub repository).

Dataset: <https://affective-meld.github.io/>

1 Introduction

Self-supervised speech models such as wav2vec 2.0 and HuBERT learn powerful representations from raw audio and are widely used as backbones for downstream speech and paralinguistic tasks [1, 2]. Their main limitation is opacity: a classifier may predict an emotion label without revealing whether the decision depends on pitch, energy, pauses, spectral structure, or other interpretable properties of speech. This is problematic for affective computing and human-facing speech analysis, where a prediction is more useful when accompanied by a meaningful explanation.

The goal of this work is therefore not to maximize state-of-the-art MELD accuracy, but to evaluate the trade-off between predictive performance and interpretability. The project uses MELD as a benchmark conversational emotion dataset [4]. A direct HuBERT classifier is compared with a Concept Bottleneck Model (CBM) [3], which forces predictions to pass through acoustic proxy concepts before the final emotion classifier. This makes the intermediate representation inspectable, rankable with SHAP [5], and editable through concept interventions.

2 Related Work and Research Gaps

Post-hoc explainability explains a trained model after prediction. For speech and audio, common approaches include SHAP over engineered features, gradient or relevance maps over waveform or spectrogram inputs, attention-based temporal weights, and perturbation methods such as audioLIME [10, 8, 9, 11]. These methods are flexible, but the explanation is not part of the predictive architecture.

Explainable-by-design methods instead embed interpretability into the model. CBMs predict human-understandable concepts before the final label, enabling inspection and intervention [3]. Concept Embedding Models relax the strict bottleneck while retaining concept-level explanations [6]. In speech, however, concepts are harder to define than objects in vision or tokens in text. A useful explanation must be grounded in time-frequency and paralinguistic properties such as pitch, intensity, pauses, rhythm, MFCCs, or voice quality. Descriptor sets such as GeMAPS/eGeMAPS motivate the use of prosodic, spectral, and cepstral concepts [7].

This project addresses three research gaps. First, strong self-supervised speech backbones are often used as black-box classifiers, with limited insight into the acoustic factors driving predictions. Second, there is no universally accepted concept taxonomy for speech emotion, so practical systems often rely on imperfect acoustic proxies. Third, many XAI pipelines show importance plots but do not test whether explanations are faithful or actionable. The proposed pipeline therefore combines a black-box baseline, a CBM, SHAP concept importance, ablation-based faithfulness, and targeted concept interventions.

3 Methodology and Implementation

The task is seven-class emotion classification on MELD utterances: anger, disgust, fear, joy, neutral, sadness, and surprise. The input modality is audio only. Videos are converted to 16 kHz waveform files; missing and failed media are logged. In the final usable set, 13,706 utterances are available. The official train, development, and test splits are preserved.

A frozen facebook/hubert-base-ls960 model extracts a 768-dimensional utterance embedding. Frame-level hidden states are pooled into one utterance-level vector. The direct black-box baseline trains an emotion classifier on this HuBERT embedding. The CBM uses the same HuBERT embedding but changes the prediction path: the embedding first predicts a 39-dimensional acoustic concept vector, and the

Table 1: Implemented experimental setting.

Component	Choice
Dataset	MELD emotion utterances
Input modality	Audio only
Backbone	Frozen HuBERT base
Embedding size	768 dimensions
Concepts	39 acoustic proxy concepts
Task	7-class emotion classification
Explanation tools	SHAP, ablation, interventions
Primary metric	Macro-F1

final classifier predicts emotion from the predicted concepts. An oracle concept classifier, trained directly on the extracted acoustic concepts, is included as a diagnostic reference for the concept set rather than as a deployable model.

The acoustic proxy concepts include duration, energy mean and variance, pause ratio, zero-crossing rate, spectral centroid, rolloff, bandwidth, pitch/voicing proxies, voiced-ratio proxy, and MFCC statistics. These concepts are interpretable at the signal-processing level, but they are not human-annotated psychological labels. Concept prediction is evaluated with R^2 and scaled error, while final classification is evaluated with accuracy, macro-F1, weighted-F1, macro-precision, and macro-recall. Macro-F1 is treated as the primary metric because MELD is strongly imbalanced and a majority-class baseline can obtain high accuracy while failing on minority emotions.

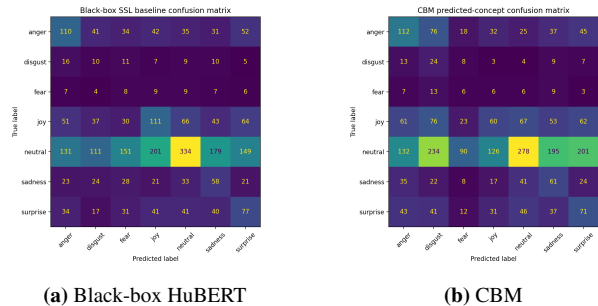
The implementation is designed to be reproducible and computationally manageable. Audio conversion, extracted acoustic concepts, and HuBERT embeddings are cached explicitly so that the final statistical analysis can be rerun without repeating the slow feature extraction stage. Failed conversions and missing media are stored as output files instead of being silently ignored. This is essential for reproducible explainability evaluation: SHAP explanations, ablation results, and model comparisons are only valid when the evaluated samples are explicitly documented and kept consistent across the black-box, CBM, and oracle models.

4 Results and Analysis

Table 2 reports test-set performance with 95% bootstrap confidence intervals from 1,000 resamples for accuracy and macro-F1. The majority baseline always predicts the neutral class. It reaches 0.481 accuracy but only 0.093 macro-F1, confirming that accuracy alone is misleading and justifying macro-F1 as the main metric. The black-box HuBERT model obtains the best macro-F1, while the CBM sacrifices performance for interpretability. A paired bootstrap gap test shows that the black-box model is distinguishable from the CBM for both accuracy and macro-F1. The black-box-CBM gaps are 0.0368 in accuracy with 95% CI [0.0149, 0.0571] and 0.0212 in macro-F1 with 95% CI [0.0020, 0.0411]. The CBM-oracle gap is not distinguishable from zero: the CBM-oracle CIs are [-0.0057, 0.0303] for accuracy and [-0.0082, 0.0227] for macro-F1. This is important because the oracle should structurally be at least competitive: it uses real extracted concepts instead of predicted

Table 2: Predictive performance on MELD test utterances. Brackets show 95% bootstrap CIs for accuracy and macro-F1.

Model	Accuracy	Macro-F1	W-F1
Majority neutral	0.481 [0.462,0.500]	0.093 [0.090,0.095]	0.313
Black-box HuBERT	0.271 [0.254,0.287]	0.214 [0.198,0.229]	0.306
CBM	0.234 [0.218,0.251]	0.192 [0.177,0.207]	0.263
Oracle concepts	0.222 [0.207,0.237]	0.185 [0.171,0.200]	0.254

**Figure 1:** Confusion matrices on the MELD test split. Both audio-only models confuse minority emotions and many samples involving the neutral class.

concepts. The fact that it is not clearly better suggests that the concept set itself is weak for MELD emotion recognition and that the small ordering difference is within split-level noise.

The absolute scores are low, but this should be interpreted in context. MELD is multimodal and conversational: many labels depend on lexical content, facial expression, speaker identity, and dialogue history. The experiment deliberately studies audio-only explainability with a frozen backbone, not a state-of-the-art multimodal MELD system. The confusion matrices in Figure 1 show that both models struggle with rare emotions such as fear and disgust. The black-box model preserves neutral and joy somewhat better, while the CBM redistributes more predictions across minority classes.

The concept prediction stage is partly successful. Several low-level acoustic concepts are recoverable from HuBERT embeddings, including duration, MFCC variability, and pitch/voicing proxies. Table 3 reports the strongest concept-prediction results. This shows that the SSL embedding contains acoustic information, but it does not imply that those concepts are sufficient for emotion recognition. The weak oracle concept classifier confirms that recoverable acoustic descriptors are only partially aligned with MELD emotion labels.

Figure 2 shows the highest concept-level SHAP importances for the CBM. The ranking is dominated by spectral and cepstral variables, especially spectral centroid, spectral rolloff, spectral bandwidth, MFCC means, and pitch/voicing proxies. This is plausible for paralinguistic analysis because spectral shape, voicing, and cepstral descriptors are common acoustic correlates of affect. However, they remain proxy concepts rather than validated labels such as “angry tone”, “breathy voice”, or “fast speech”.

Faithfulness was tested by ablating concepts selected by

Table 3: Most predictable acoustic proxy concepts from HuBERT embeddings.

Concept	Test R^2
Duration	0.901
MFCC 1 standard deviation	0.853
Pitch/voicing proxy std.	0.804
MFCC 2 standard deviation	0.763
MFCC 3 standard deviation	0.689
Spectral bandwidth mean	0.533
Pitch/voicing proxy mean	0.500

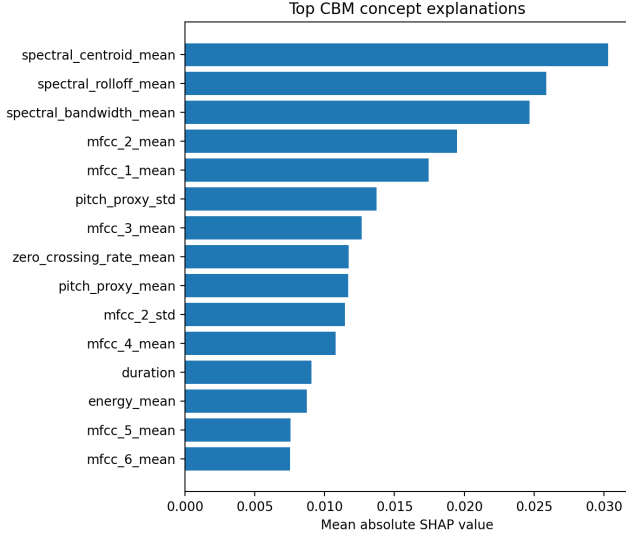


Figure 2: Top concept-level SHAP importances for the CBM. Spectral and cepstral concepts dominate the explanation.

SHAP and comparing the performance drop with random concept ablations. Table 4 reports the updated random-ablation standard deviations over 20 runs. The strongest evidence appears at top-5, where SHAP-selected concepts cause a larger accuracy and macro-F1 drop than random concepts. The evidence is not uniform: top-1 and top-10 are less consistent. The correct conclusion is therefore partial faithfulness, not full validation of the SHAP ranking.

Targeted interventions further illustrate the practical value of the bottleneck. For selected test examples, increasing or decreasing one standardized concept while keeping the others fixed can change the predicted class or the probability of the original class. This does not create a physically valid edited audio signal; rather, it evaluates whether the model’s decision is causally sensitive to an interpretable internal variable. Such concept-space editing is available in the CBM but not in the direct HuBERT baseline.

Overall, the results match the expected accuracy-interpretability trade-off. The direct HuBERT classifier has access to the full latent representation, so it retains more predictive information. The CBM compresses the decision through a limited set of acoustic variables, which reduces performance but makes the prediction pathway inspectable. In a high-

Table 4: Faithfulness by concept ablation. Random rows are mean $\pm$ std over 20 runs.

Ablation	Acc. drop	Macro-F1 drop
Top-1 SHAP	-0.005	0.014
Random-1	0.002 $\pm$ 0.015	0.002 $\pm$ 0.007
Top-3 SHAP	0.009	0.022
Random-3	0.015 $\pm$ 0.018	0.010 $\pm$ 0.009
Top-5 SHAP	0.042	0.038
Random-5	0.018 $\pm$ 0.021	0.019 $\pm$ 0.011
Top-10 SHAP	0.001	0.025
Random-10	0.033 $\pm$ 0.021	0.028 $\pm$ 0.012

stakes deployment, this trade-off would only be acceptable if the concepts were validated and useful to domain users. For this project, the main contribution is to demonstrate the complete explainable-by-design workflow and to show empirically where the current concept design succeeds and where it fails.

5 Limitations and Threats to Validity

The main limitation is that the concepts are automatically extracted acoustic proxies, not expert annotations. They are interpretable in a signal-processing sense, but they should not be treated as verified psychological or clinical concepts. Pitch-related variables are fast voicing proxies rather than full fundamental-frequency tracks, and pause ratio depends on an energy threshold.

A second limitation is the audio-only setting. MELD labels often require utterance text, speaker context, facial expression, and dialogue history. This partly explains the low absolute scores and the weakness of the oracle concept classifier. A third limitation is the frozen-backbone design. Freezing HuBERT makes the experiment computationally feasible and isolates the concept layer, but fine-tuning or multimodal modeling would likely improve performance. Finally, explanation evaluation is model-level rather than human-grounded: ablation and interventions test faithfulness and actionability inside the classifier, but no listener or speech-expert study is included. For this reason, the claims should be limited to transparency, diagnostic usefulness, and internal faithfulness. The work should not be interpreted as proving that the explanations are psychologically meaningful to humans or sufficient for clinical use.

6 Conclusion

This project implemented and evaluated an explainable-by-design pipeline for speech emotion analysis. A frozen HuBERT black-box classifier was compared with a CBM over 39 acoustic proxy concepts. The black-box model achieved higher predictive performance, but the CBM provided concept-level explanations, faithfulness analysis, and intervention capability. The results support the usefulness of concept bottlenecks for transparency, while showing that automatically extracted acoustic proxies are not yet strong enough to fully explain MELD emotion labels. Future work should use a richer prosody taxonomy, standardized eGeMAPS descriptors, human concept annotations, and multimodal text-audio context.

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